Comparative Effects of Credit Utilization and Credit Limit Adjustments on Predicted Default Risk

Sree Prabhav Bandakavi
University of Virginia
School of Data Science
Charlottesville, Virginia
kqy9hc@virginia.edu

Humaid Billoo
University of Virginia
School of Data Science
Charlottesville, Virginia
ske6pe@virginia.edu

Claudio Cela
University of Virginia
School of Data Science
Charlottesville, Virginia
cc5hu@virginia.edu

Jack Thompson Hays
University of Virginia
School of Data Science
Charlottesville, Virginia
jth2uyr@virginia.edu

I. Abstract:

This paper investigates how targeted financial interventions, specifically reducing credit utilization and increasing credit limits, affect the predicted probability of credit card default under machine learning models trained on the UCI Default of Credit Card Clients dataset. The dataset contains 30,000 credit card holders in Taiwan and includes demographic, repayment history, billing, payment, and credit limit information. Five classification models were developed and compared: Logistic Regression, Support Vector Machine, Random Forest, XGBoost, and a Multilayer Perceptron. Each model was trained using a stratified 70/20/10 train-validation-test split. After engineering six monthly credit utilization ratios, the dataset contained 29 raw input features, which expanded to 97 model input features after one-hot encoding categorical repayment status, education, marital status, and sex variables.

To assess model robustness, the final modeling pipeline was evaluated across three random seeds. The multi-seed results showed that the strongest nonlinear models performed comparably, with Random Forest achieving an average AUC-ROC of 0.777 $\pm$ 0.012 and F1 score of 0.537 $\pm$ 0.016, XGBoost achieving an average AUC-ROC of 0.778 $\pm$ 0.009 and F1 score of 0.537 $\pm$ 0.019, and the MLP achieving an average AUC-ROC of 0.777 $\pm$ 0.012 and F1 score of 0.528 $\pm$ 0.023. These results indicate that no single nonlinear model clearly dominated across seeds, while all three models substantially outperformed the linear baseline.

SHAP analysis showed that repayment status variables were the dominant predictors of default across models, while credit limit and utilization-related features contributed weaker but still meaningful predictive signal. Counterfactual simulations were then conducted to compare three intervention scenarios: reducing bill amounts while holding credit limits constant, increasing credit limits while holding bill amounts constant, and increasing both credit limits and bill amounts proportionally to isolate the pure credit limit effect while holding utilization constant.

The results indicate that interventions targeting both credit utilization and credit limits have the potential to alter the default probability of loans in the model. The models exhibit similar responses to interventions targeting credit utilization versus credit limits, indicating that each of these variables is a contributor to the prediction of loan default probability. Furthermore, the responses of the models differ according to the risk of the loan's clients, with the high-risk clients exhibiting larger decreases in their predicted default probability in response to interventions that target their credit utilization. These results only indicate the behavior of the models in response to the interventions, and do not indicate any causal relationship between those interventions and the variable of interest. Overall, this study demonstrates how predictive modeling, interpretability methods, robustness testing, and counterfactual analysis can be combined to evaluate how financial interventions may affect modeled credit default risk. Our code and work description are available on GitHub https://github.com/Group-10-Deep-Learning-Project/G10— Comparative-Effects-of-Credit-Utilization-on-PredictedDefault-Risk.

II. Introduction:

Credit risk models are commonly used to estimate the probability of borrower default using financial and behavioral features such as repayment history, billing amounts, and credit limits. For financial institutions, this often involves the use of credit scores. Credit scores are a traditional scoring methodology used to evaluate a client's creditworthiness and potential risk, which typically translates into different credit terms or interest rates depending on the score. Although machine learning methods have improved the accuracy of default risk prediction, they are primarily used for classification and ranking rather than to evaluate strategies that could reduce the predicted probability of default. In practice, lenders may attempt to mitigate risk either by reducing borrower credit utilization, for example, by encouraging higher payments, or by adjusting credit limits as an intervention strategy. However, it remains unclear which of these approaches produces a greater reduction in the predicted default risk. This study seeks to address this gap by developing and comparing multiple machine learning models to predict borrower default probability using all available financial and behavioral variables, evaluating their performance on out-of-sample test data to identify the most accurate model, and applying SHAP analysis to quantify the relative influence of each parameter on predicted default risk. Specifically, this study simulates three targeted interventions: reducing client bill amounts while holding credit limits constant, increasing credit limits while holding bill amounts constant, and increasing both credit limits and bill amounts proportionally while holding utilization constant. This paper aims to investigate the effect that specific interventions have upon the predicted default probabilities of borrowers. Beyond building the models that calculate these default probabilities, the goal of this project is also to gain insights from the models that can be applied to financial decision-making.

III. Dataset:

The Default of Credit Card Clients dataset was obtained from the UCI Machine Learning Repository and contains 30,000 observations of individual credit card clients in Taiwan with 23 explanatory variables and a binary default indicator. [7] The dataset has a default rate of 22.12% (6,636 defaulters, 23,364 non-defaulters) and we verified zero missing values across all features. Features include demographic attributes (sex, education, marital status, age), credit limits in Taiwan dollars (TWD) ranging from 10,000 to 1,000,000 across clients (mean 167,484), repayment status over six months (September–April 2005), monthly bill statements (X12–X17), and payment amounts (X18–X23). We verified that bill statements range from -165,580 to 964,511 across all clients and months, with negative values (approximately 2% of observations) representing credits or refunds. Payment amounts range from 0 to 873,552 across all clients and months, with zero payments (no payment made) occurring in 17% to 24% of observations depending on the month.

The most significant data quality issue involves repayment status variables (X6–X11). UCI documentation specifies values of -1 (paid in full) and 1–9 (months delayed), but we verified the actual data contains undocumented values -2 and 0 that together constitute 58–65% of observations each month. Value 0 (ranging from 49% to 56% across the six months), the single most common state yet absent from documentation and likely represents "revolving credit" (minimum payment made, balance carried forward). Value -2 (ranging from 9% to 16% across the six months) likely represents "no consumption" (unused credit line). To understand the relationship between repayment status and default risk, we calculated default rates for each status category using September 2005 as a representative month (Table I). The patterns observed in September are consistent across all six months in the dataset. Default rates represent the percentage of clients within each repayment status category who eventually defaulted. For example, of the 2,759 clients (9.2% of 30,000) with status -2 in September 2005, 298 (10.8%) defaulted on payment in the following month. We verified that default rates do not follow the numeric ordering. States -2 and -1 show similar low risk (10.8% and 13.2%), state 0 shows moderate risk (20.1%), then risk more than doubles to 41.7% at state 1. This nonlinear pattern indicates that the encoding does not represent sequential ordering, and rather each state reflects qualitatively different payment behavior.

TABLE I: Default Rate by Repayment Status (September 2005)

Value	Interpretation	% of Data	Default Rate
-2	No consumption	9.2%	10.8%
-1	Paid in full	19.0%	13.2%
0	Revolving credit	49.1%	20.1%
1	1 month delayed	12.3%	41.7%
2	2 months delayed	8.9%	63.0%
3+	3+ months delayed	1.6%	80.4%

Note. Values -2 and 0 are undocumented in the original UCI dataset description.

Given these findings, we treat X6–X11 as categorical variables using one-hot encoding allowing models to learn true relationships without false sequential assumptions. Education (X3) and marital status (X4), which also contain undocumented values, receive categorical encoding.

As part of the feature engineering process, a credit utilization ratio was computed to explicitly capture the proportion of available credit being used by each client. Credit utilization is defined as the ratio of the monthly bill amount to the total credit limit and is calculated for each of the six observed months as

$$\text{Utilization}_t = \frac{\text{BILL\_AMT}_t}{\text{LIMIT\_BAL}}, \quad t = 1, 2, \ldots, 6.$$

In the dataset, the credit limit is represented by column X_1 (LIMIT_BAL), while the monthly bill statement amounts are represented by columns X_{12} through X_{17} (BILL_AMT1–BILL_AMT6). This transformation produces six utilization features that more directly measure borrower credit usage relative to their available credit. These utilization variables are denoted as C_{24} through C_{29}, with the six-month average utilization denoted as C_{30} when included in the engineered feature set.

After feature engineering, the final raw feature set contains the original explanatory variables plus six monthly utilization ratios. These raw variables are then transformed during preprocessing through one-hot encoding of categorical variables, including repayment status, education, marital status, and sex. This produces a final model input matrix with 97 encoded features. All models use this same 97-feature representation, ensuring that performance differences reflect model behavior rather than differences in input dimensionality.

IV. Literature Survey:

The study by Umi Munirah et al. applies classic machine learning classification techniques to predict credit card debt default using data obtained from the UCI Machine Learning Repository. The preprocessing phase included handling missing values and applying feature selection methods such as forward selection, backward elimination, and an optimized wrapper approach. The dataset was divided into a 70:30 ratio for training and testing, respectively, and three models were implemented: Decision Tree, Logistic Regression, and Naive

Bayes. Model performance was evaluated using confusion matrix metrics, including accuracy, sensitivity, specificity, precision, and area under the ROC curve. The results indicated that a Decision Tree using the Gini Index with forward selection yielded the strongest performance. However, the study has several notable gaps. It relies solely on traditional models and does not explore more advanced approaches such as Random Forests, Gradient Boosting, or neural networks, which have demonstrated strong performance in credit risk prediction studies. Additionally, the paper does not address class imbalance or incorporate cross-validation, both of which could improve predictive performance and model generalizability. [4]

Building on this, recent research in credit risk modeling has focused on improving predictive performance using machine learning methods such as Logistic Regression, Random Forests, Gradient Boosting Machines, and Deep Neural Networks. These models commonly use demographic attributes, repayment history, billing amounts, payment behavior, and credit limits to estimate the probability of default. Model performance is typically evaluated using metrics such as AUC, accuracy, and log loss. Tree-based models introduce nonlinear partitioning of the feature space, allowing interactions between repayment status, billing amounts, payment amounts, and credit limits to be captured more effectively. Random Forests and Gradient Boosting methods can further improve performance by aggregating multiple weak learners and reducing overfitting through bagging and boosting strategies. More recently, deep neural networks have been applied to credit scoring tasks to learn high-dimensional nonlinear representations of borrower behavior. However, most existing studies focus primarily on classification performance and do not examine how targeted modifications to key financial variables influence predicted default probability. As a result, a gap remains in understanding whether reducing credit utilization through balance reduction or increasing credit limits produces greater marginal reductions in modeled default risk under machine learning-based credit risk models. [1]

Furthermore, the paper by Anas Arram et al. examined the effectiveness of different machine learning methods for credit score prediction. The study used multiple preprocessing methods before comparing several models, including MLP, XGBoost, and LightGBM. The authors found that LightGBM achieved the highest accuracy, while MLP was recommended because it produced fewer false negatives. This distinction is important in credit risk contexts because false negatives may represent borrowers predicted as low risk who later default. However, one limitation of the study is the relatively small sample size of 500 observations, which limits the generalizability of the results and provides further motivation for evaluating credit risk models on larger datasets. [3]

While much of the literature emphasizes predictive modeling, fewer studies examine operational decision-making interventions. Pestana and Areyan propose a machine learning approach to automate credit card limit adjustment decisions (CLAD) at a Venezuelan bank, where the growing number of credit card holders made manual review by committee increasingly infeasible. Using a dataset of 10,000 real-world CLAD decisions collected over five years, the study trains and compares neural network and XGBoost models with cost-sensitive learning. The authors use grid search with 10-fold cross-validation to tune hyperparameters and frame misclassification costs asymmetrically. In this setup, false positives carry a higher cost than false negatives because a false positive means granting a limit increase to a client who later defaults. This makes cost minimization, rather than accuracy alone, the primary optimization goal. The XGBoost model was ultimately selected as the proposed solution, achieving 94.91% accuracy and lower total cost than the neural network while also providing greater interpretability. Agreement between the model and the human committee was evaluated using Cohen's kappa coefficient, which returned $\kappa = 0.81$, indicating almost perfect agreement. However, the study focuses primarily on whether to grant a credit limit increase and does not examine the downstream effect that limit adjustments have on predicted default probability relative to changes in credit utilization. As a result, a gap remains in understanding whether credit limit adjustments or balance reductions produce greater reductions in modeled default risk, which this study directly addresses. [6]

In summary, the existing literature demonstrates significant progress in improving credit default prediction through machine learning. However, most prior work focuses on classification performance rather than comparing the modeled effects of possible financial interventions. This study addresses that gap by evaluating how predicted default probability changes under balance reduction, credit limit increase, and utilization-controlled credit limit intervention scenarios.

V. METHOD:

This study constructed a classification pipeline to predict borrower default probability. The dataset was first verified for missing values and class balance, confirming a default rate of approximately 22%. Six credit utilization ratios were engineered using $\text{Utilization}_t = \text{BILL\_AMT}_t/\text{LIMIT\_BAL}$ for each observed month. After feature engineering, the dataset contained 29 raw input features. These raw variables were then transformed through one-hot encoding of categorical variables, producing a final model input matrix with 97 features.

To ensure methodological rigor, we verified that no data leakage was introduced during feature engineering. The utilization features $\text{UTIL}_t = \text{BILL\_AMT}_t/\text{LIMIT\_BAL}$ for $t = 1, \ldots, 6$ were constructed solely from information available prior to the prediction window, while the target variable corresponds to default status in the following month. Therefore, no future information was used in feature construction. Additionally, any feature scaling applied for models such as Logistic Regression, SVM, or MLP was fit exclusively on the training data and then applied to the validation and test sets to prevent information leakage.

The data was split into training (70%), validation (20%), and test (10%) sets using a randomized stratified split to preserve class balance across all three partitions.

Five classification models were trained and compared on the same training, validation, and test splits:

1) **Logistic Regression:** An L2-regularized logistic regression model trained on the 97-feature encoded input matrix served as the non-deep-learning baseline, providing a performance benchmark for the subsequent nonlinear models.
2) **SVM (Support Vector Machine):** An SVM classifier with a radial basis function (RBF) kernel was implemented as an additional baseline model. Hyperparameters were tuned using cross-validation on the training data.
3) **Random Forest:** A tree-based ensemble model was trained to capture nonlinear feature interactions through bagging. The Random Forest model used the same 97-feature encoded input matrix as the other models and was evaluated to determine whether an ensemble method offered gains over the logistic regression baseline.
4) **XGBoost:** A gradient boosting model was trained and evaluated because prior work has shown that boosted tree models perform strongly on credit risk prediction tasks. XGBoost was used as a high-performing nonlinear benchmark against the Random Forest and MLP models.
5) **MLP (Multilayer Perceptron):** A neural network was trained using the 97-feature encoded input matrix. The architecture included an input layer, a hidden layer of 128 units with ReLU activation, BatchNorm, and Dropout of 0.3, followed by a second hidden layer of 64 units with ReLU activation, BatchNorm, and Dropout of 0.3, and a final output layer with a Sigmoid activation. The model used weighted binary cross-entropy loss, the AdamW optimizer, and early stopping.

The repayment status variables (PAY_0 through PAY_6) use numeric codes where values such as -2 (no consumption) and -1 (paid in full) represent distinct categories rather than a meaningful numeric order. To prevent the models from incorrectly treating these values as ordered, repayment status variables were converted into separate binary columns using one-hot encoding before model training. Education, marital status, and sex were also one-hot encoded.

The final model comparison was conducted across three random seeds (42, 123, and 456). For each seed, models were trained on the training split, tuned using the validation split, and evaluated on the held-out test split. Random Forest was selected for the main counterfactual analysis because it achieved performance comparable to XGBoost and MLP while also supporting interpretable SHAP-based feature analysis.

VI. Results

A. Classification Performance

Table II reports the multi-seed classification performance across the five models. Across three random seeds, Random Forest, XGBoost, and MLP performed similarly, with AUC-ROC values of 0.777, 0.778, and 0.777, respectively. Random Forest and XGBoost also achieved nearly identical F1 scores of 0.537 and 0.536. Logistic Regression achieved lower but still competitive performance, while SVM had the lowest AUC-ROC among the five models. Overall, these results indicate that the nonlinear models performed comparably and that no single model clearly dominated across seeds.

TABLE II: Classification Model Performance on Test Set (Mean ± Standard Deviation Across 3 Seeds)

Model	AUC	F1	Accuracy
Logistic Regression	0.763 ± 0.012	0.521 ± 0.007	0.787 ± 0.015
SVM	0.726 ± 0.008	0.489 ± 0.020	0.793 ± 0.009
XGBoost	0.778 ± 0.009	0.536 ± 0.012	0.783 ± 0.008
Random Forest	0.777 ± 0.012	0.537 ± 0.015	0.783 ± 0.010
MLP	0.777 ± 0.012	0.528 ± 0.023	0.769 ± 0.018

Model	Precision	Recall
Logistic Regression	0.522 ± 0.035	0.525 ± 0.047
SVM	0.542 ± 0.030	0.450 ± 0.056
XGBoost	0.510 ± 0.018	0.566 ± 0.032
Random Forest	0.510 ± 0.020	0.568 ± 0.034
MLP	0.482 ± 0.030	0.584 ± 0.013

Note. Results represent the mean and standard deviation across three random seeds (42, 123, 456) using a stratified 70/20/10 train/validation/test split with one-hot encoding applied to repayment status, education, marital status, and sex variables.

B. SHAP Feature Importance

TABLE III: Average Top SHAP Feature Rankings Across Seeds

Rank	SVM	RF	XGBoost	MLP
1	Feature_35	PAY_0_0	PAY_0_2	Feature_35
2	Feature_46	PAY_0_2	PAY_2_2	Feature_37
3	Feature_0	PAY_2_2	PAY_3_2	Feature_34
4	Feature_37	PAY_3_2	PAY_4_2	Feature_36
5	Feature_89	PAY_AMT1	PAY_AMT1	Feature_45

Note. Rankings for SVM and MLP were averaged across three random seed runs using SHAP feature importance plots. Random Forest and XGBoost rankings were retained from their stable gain/SHAP importance outputs. Features labeled as "Feature_X" correspond to one-hot encoded repayment-status, payment-history, and engineered utilization variables generated during pre-processing. Across models, repayment behavior and credit utilization variables consistently exhibited the strongest influence on predicted default probability.

C. Counterfactual Intervention Analysis

Table IV reports the mean absolute change in predicted default probability for the Random Forest model across the three intervention levels. Intervention A simulates reducing clients' bill balances while holding credit limits constant. Intervention B simulates increasing clients' credit limits while holding bill amounts constant. Intervention C increases both credit limits and bill amounts proportionally so that utilization remains approximately constant.

Across the Random Forest model, intervention response magnitudes remained relatively stable across the 10%, 25%, and 50% intervention levels. Intervention C produced the largest response magnitude at the 50% intervention level, while Intervention B produced the smallest response magnitude

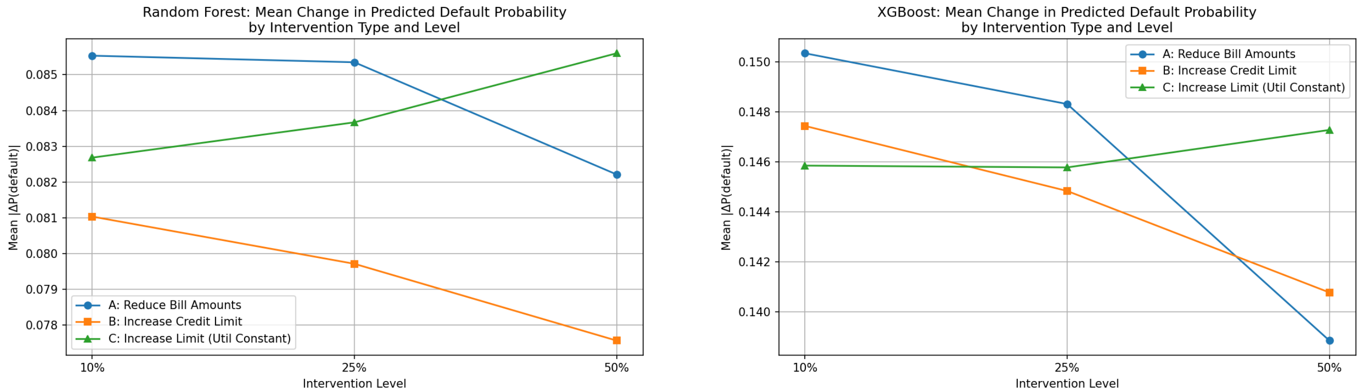

Fig. 1: Counterfactual intervention analysis across the 3,000-client test set. Left: Random Forest mean $|\Delta P(\text{default})|$ by intervention type and magnitude. Right: XGBoost mean $|\Delta P(\text{default})|$ by intervention type and magnitude. Both models showed similar response patterns across intervention types, though Random Forest exhibited slightly more stable response magnitudes across intervention levels compared with XGBoost.

across all three levels. This suggests that changing credit limits alone had less impact on the model's predicted default probabilities than changing bill balances or changing limits while maintaining utilization. The XGBoost model showed a similar pattern across intervention types, though its response magnitudes were slightly less stable across the three intervention levels compared with Random Forest, which is why we decided to stick with that as our main model.

TABLE IV: Counterfactual Intervention Results: Mean $|\Delta P(\text{default})|$ by Intervention Type and Magnitude (Random Forest)

Intervention	**10%**	**25%**
A: Reduce Bills	0.078 ± 0.018	0.077 ± 0.018
B: Increase Limit	0.076 ± 0.016	0.075 ± 0.016
C: Limit Inc., Util. Constant	0.078 ± 0.017	0.080 ± 0.018

Intervention	**50%**
A: Reduce Bills	0.073 ± 0.017
B: Increase Limit	0.073 ± 0.016
C: Limit Inc., Util. Constant	0.082 ± 0.019

Note. Values are reported as mean $\pm$ standard deviation across three random seeds. Intervention A reduces bill balances while holding credit limits constant. Intervention B increases credit limits while holding bill amounts constant. Intervention C increases both credit limits and bill amounts proportionally so utilization remains approximately constant. Because absolute values are reported, larger values indicate larger model response magnitude, not necessarily larger reductions in predicted default probability.

D. Client Segmentation

Table V reports the directional change in predicted default probability at the 25% intervention level for high-risk and low-risk clients under the Random Forest model. High-risk clients are defined as those with baseline predicted default probability above 0.5, while low-risk clients are defined as those with baseline predicted default probability below 0.4.

For high-risk clients ($n = 640$), both interventions reduced predicted default probability on average. Intervention A produced a larger average reduction (-0.0636) than Intervention B (-0.0530), suggesting that the Random Forest model responds more favorably to bill-balance reductions for borrowers already classified as high risk. For low-risk clients ($n = 1894$), both interventions increased predicted default probability on average, with Intervention B producing a larger increase (0.0510) than Intervention A (0.0401). This result indicates that the same intervention can affect model predictions differently depending on the client's baseline risk profile.

TABLE V: Segmentation Analysis: Mean $\Delta P(\text{default})$ at 25% Intervention Level by Risk Group (Random Forest)

Group	**n**	**Baseline P**	**Int. A**	**Int. B**
High-Risk	640	> 0.5	-0.0636	-0.0530
Low-Risk	1894	< 0.4	0.0401	0.0510

Note. High-risk clients are those with a baseline predicted default probability of above 0.5. Clients with a predicted default probability below 0.4 are considered to be of low risk. The remaining 466 clients fall somewhere in between these two risk categories and are, therefore, excluded from the analysis. Negative values in the tables indicate a reduction in the predicted default probability for those clients, while positive values indicate an increase in the default probability. For clients who are determined to be high risk, Intervention A results in a larger reduction in their predicted default probability than does Intervention B. For clients who are of low risk, however, both interventions result in an increase in default probability, though Intervention B leads to a larger increase than Intervention A.

E. Causal Limitations

The counterfactual results reflect model behavioral analysis rather than causal inference. Machine learning models learn correlations rather than true causal relationships, and simulating a credit limit increase for a high-risk client may artificially reduce predicted default probability in ways that would not reflect real-world outcomes. Results should be interpreted as describing how the trained model responds to feature disturbances rather than as narrow financial policy recommendations.

VII. Conclusion

The purpose of this research was to determine the impact that financial interventions have on the predicted risk of individuals of defaulting on their credit cards using the UCI Default of Credit Card Clients dataset. Five different machine learning classification models were trained and evaluated: Logistic Regression, SVM, Random Forest, XGBoost, and MLP. Random Forest and XGBoost models both exhibited the strongest performance overall, with Random Forest achieving an average AUC-ROC value of 0.777 and XGBoost achieving an average AUC-ROC value of 0.778. Both of these scores indicate that these models outperform the Logistic Regression model, which indicates that the nonlinear components of the data may play a role in the predictions of defaulting on credit cards.

While the XGBoost model exhibited slightly stronger performances in some evaluations of the models, the Random Forest model exhibited comparable performance to the XGBoost algorithm while also providing individuals with a more interpretable analysis of the features of the data. So, the Random Forest model was selected for analysis of counterfactual outcomes.

The analysis of the importance of each feature within the Random Forest and XGBoost models indicated that the model predicted that the repayment status of the individuals is the strongest indicator of whether they will default on their credit cards, followed by features related to the credit limits and the amount of credit that is being utilized by those individuals. As such, the findings of the model help to explain the reason that interventions related to credit limits and credit utilization have any effect on the default rates of those individuals.

Through conducting counterfactual analyses using the Random Forest model, it becomes indicated that implementing interventions to the bill balances and credit limits of the borrowers results in changes to the predicted probability of those borrowers defaulting on their loans. Each of these interventions had different magnitudes of change in relation to the model, indicating the sensitivity of the model to changes to those features. Intervention A and Intervention B had similar magnitudes of change to the predicted default probability for those borrowers, however Intervention C indicated that even without altering the bill balances of those borrowers, altering their credit limits can have an impact on the predicted probability of default. As such, these results indicate that the machine learning models are sensitive to changes to both the utilization and credit limit of the borrowers of those loans.

The segmentation analysis was the most important finding from this research. For those borrowers with high risk classifications, it was more effective to reduce their credit card balances than to increase their limits. For those borrowers with low risk classifications, while both options had a positive mean change to the default probability, increasing their limits resulted in the larger change to that probability. As such, the model exhibits different responses to the same interventions according to the risk classification of the borrower.

Overall, this project helped to indicate how each of these machine learning models can be used not just to determine the risk of individuals of defaulting on their loans, but also how those models may respond to various financial interventions. However, these findings are not indicative of the causal relationship between those financial interventions and the defaulting of loans by those individuals. Instead, they are based on the behavior of the models with data that was created using historical data of individual borrowers of those loans. Due to this inherent relationship between the models and the borrowed loans data, future work may extend this project to include additional methods of performing causal analyses, using additional borrowed loan data, and performing additional analyses on additional datasets of borrowers of credit cards.

VIII. Member Contributions

Sree Prabhav Bandakavi implemented the SVM (Support Vector Machine) and Logistic Regression models. Humaid Billoo implemented and analyzed the XGBoost model, as well as performed analyses of the dataset and its validation. Claudio Cela implemented the Random Forest model and contributed to the development of the methodology behind the models. Jack Thompson Hays implemented the MLP (Multilayer Perceptron) model and worked with Claudio on the data processing and analysis.

References

[1] Stefania Albanesi and Domonkos F. Vamossy. Predicting consumer default: A deep learning approach, 2019.
[2] Anthropic. Claude 3.5 sonnet, 2025.
[3] Anas Arram, Masri Ayob, Musatafa Abbas Abbood Albadr, Alaa Sulaiman, and Dheeb Albashish. Credit card score prediction using machine learning models: A new dataset, 2023.
[4] Umi Munirah Ishak, Nur Nabilah Arina Ali, and Norshahida Shaadan. Machine learning-based approaches for credit card debt prediction. *MALAYSIAN JOURNAL OF COMPUTING (MJOC)*, 2024.
[5] OpenAI. Chatgpt. https://chat.openai.com, 2026.
[6] Diego Pestana and Enrique Areyan Viqueira. Automating credit card limit adjustments using machine learning, 2025.
[7] I-Cheng Yeh. Default of Credit Card Clients. UCI Machine Learning Repository, 2009. DOI: https://doi.org/10.24432/C55S3H.